



POSLink Feature Check List

DRAFT

Version: 1.06

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PAX Technology, Inc.

Date	Version	Description
2018/01/24	V1.05	Add IAD and ARC in Receipt Printing according to Tsys EMV Implementation Guide
2018/10/04	V1.06	1) Add "S500", "Q30", "D220", "E500", "A920", and "Other" in Terminal Type 2) Delete "MT30" in Terminal Type

Contents

Contents	1
1. POS Company Name	2
2. POS System.....	2
3. Communication Method Supported	3
4. Terminal Type.....	3
5. EDC Supported Trans Type.....	4
6. Additional Features	5
7. Batch Commands	6
8. Reports	6
9. Logging	7
10. Administrative Features	7
11. Mandatory Requirements	8
12. Good Practice	8
13. Exception Handling.....	9
14. Receipt Samples	10

1. POS Company Name

POS Business contact	
Name	
Phone	
Address	
POS Developer	
Name	
Phone	
Address	
POS Support contact	
Name	
Phone	
Address	

2. POS System

POS System	
POS System Name	
Version	
Date	
POS Type	☐ Windows (Native appl) ☐ Android (Native appl) ☐ Linux (Native appl) ☐ Standalone ECR, Brand_____Model_____ ☐ Web browser POS Client _____ Other OS
POS Development Language	☐ C/C++

POS System	
	☐ Java _____ i.e. VB6, C# etc...
Special Requirements	☐ .Net version _____ ☐ Security key (i.e. Hardlock) ☐ Low Level Spec _____ Other
POS Architecture	☐ Server/Client (The terminal communicates to the POS client via the Server, the POSLink is integrated on the server side) ☐ Direct (The terminal communicates directly to the POS system)
Integration Type	☐ Semi-Integration (used as Payment device) or ☐ Full Integration (used as peripheral device)

3. Communication Method Supported

Communication Type	Supported	Remark
Serial Port	☐ COM Port	Serial communication
Ethernet	☐ TCP/IP	LAN connection
	☐ HTTP	HTTP Get method/Web Browser POS system
	☐ HTTPS	
	☐ SSL	

4. Terminal Type

Terminal Type	Check List
SP30	☐
D200	☐
D210	☐
D220	☐

Terminal Type	Check List
S300	☐
S500	☐
Px5	☐
Px7	☐
E500, Q30	☐
A920	☐
Other	☐

5. EDC Supported Trans Type

Payment type and transaction type is Host dependent feature not all host application supports it.

EDC Type	Transaction Type	Check List
Credit	SALE	☐
	RETURN	☐
	FORCE	☐
	AUTH (pre auth)	☐
	POSTAUTH	☐
	VOID	☐
	Void SALE	☐
	Void RETURN	☐
	Void AUTH	☐
	Void POSTAUTH	☐
	Void FORCED	☐
	INQUIRY	☐
	ADJUST (tip adjustment)	☐
	VERIFY	☐
DEBIT	SALE	☐
	RETURN	☐
	INQUIRY	☐
	V/SALE	☐
	V/RTRN	☐
	CashBack	☐
EBT (EBT_FOODSTAMP EBT_CASHBENEFIT)	SALE	☐
	RETURN	☐
	WITHDRAWAL	☐
	INQUIRY	☐
	Void SALE	☐
	Void RETURN	☐
	Void WITHDRAWAL	☐
GIFT	SALE	☐

EDC Type	Transaction Type	Check List
	AUTH	☐
	POSTAUTH	☐
	INQUIRY	☐
	ACTIVATE	☐
	ISSUE	☐
	RELOAD	☐
	RETURN	☐
	FORCED AUTH	☐
	VOID	☐
	CASHOUT	☐
	DEACTIVATE	☐
	REPLACE	☐
	ADJUST	☐
	MERGE	☐
	REPORTLOST	☐
	FORCED ISSUE	☐
	FORCED ADD	☐
	UNLOAD	☐
	RENEW	☐
LOYALTY	REDEEM	☐
	ISSUE	☐
	RELOAD	☐
	VOID	☐
	INQUIRY	☐
	REPLACE	☐
	DEVACTIVATE	☐
	RETURN	☐
	ACTIVATE	☐
	MERGE	☐
	RENEW	☐
CASH	UNLOAD	☐
	SALE	☐
	RETURN	☐
	VOID	☐
	V/SALE	☐
	V/RTRN	☐

6. Additional Features

Value	Check List	Remark
Token	☐	Tokenization support
Level II	☐	Level II data

Leve III	☐	Level III data
SAF	☐	Store and Forward
eWIC	☐	
WEX/FLEET	☐	
FSA	☐	

7. Batch Commands

Value	Check List	Remark
BATCHCLOSE	☐	POS system should support this, otherwise the terminal need to set to autobatch.
FORCEBATCHCLOSE	☐	
BATCHCLEAR	☐	
SAFUPLOAD	☐	Upload SAF transactions
DELETESAFFILE	☐	Delete store and forward(SAF) records once upload success or failed.

8. Reports

Type	Check List	Remark
LOCALTOTALREPORT	☐	optional
LOCALDETAILREPORT	☐	optional
LOCALFAILEDREPORT	☐	optional
HOSTREPORT	☐	optional
HISTORYREPORT	☐	optional
SAFSUMMARYREPORT	☐	optional

9. Logging

Logging activity	Check List
POSLink	☐ Debug level ☐ Error level
POS System	☐ Debug Level ☐ Error Level

10. Administrative Features

Command	Check List
INIT	☐
GETVAR	☐
SETVAR	☐
SHOWDIALOG	☐
GETSIGNATURE	☐
SHOWMESSAGE	☐
CLEARMESSAGE	☐
RESET	☐
UPDATEIMAGE	☐
DOSIGNATURE	☐
DELETEIMAGE	☐
SHOWTHANKYOU	☐
REBOOT	☐
GETPINBLOCK	☐
INPUTACCOUNT	☐
RESETMSR	☐
INPUTTEXT	☐
CHECKFILE	☐
AUTHORIZECARD	☐
COMPLETEONLINEEMV	☐
REMOVECARD	☐

GETEMVTLVDATA	☐
SETEMVTLVDATA	☐
INPUTACCOUNTWITHEMV	☐
COMPLETECONTACTLESSEMV	☐
SETSAFPARAMETERS	☐
SHOWTEXTBOX	☐

11. Mandatory Requirements

The POS system must support the following features

Requirement	Remark
Partial Authorization (Split Tender)	Partial authorization is a payment processor requirement, PAX terminal certified the partial authorization.
Must check for Approved amount returned	Approval amount may be different than the request amount, you can not assume the request amount is equal to the approval amount.
Balance print on Customer Copy	Receipt requirement i.e. EBT balance Prepaid balance Gift/Loyalty balances
Terminal Error code handling	Terminal returns 6 digit error code. POS must display/print this terminal error code.
Host error code handling	Host error is depending on the payment processor, when the terminal returns "000100" result code, it means host error. The POS must display/print the host error code ("Host Code")
Receipt Printing	The following EMV tags must be printed on the receipt, see examples below. TC: TVR: AID: TSI: ATC: APPLAB: IAD: ARC:

12. Good Practice

1. Date and Time Sync, the POS should call the POSLink SetVar(dateTime) to synchronize the terminal's date and time with the POS system.

13. Exception Handling

1. Exception handling – if the POS did not get the valid POSlink object after an authorization request, the POS needs to find out what happened to the request. It can call the Local Detail Report API to find out if the request was approved or not. If the transaction was approved the terminal will return that record. The ECR reference number can be used to uniquely identify an authorization record.

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14. Receipt Samples

MSR Receipt Sample	08/28/2015	14:57:17
	CREDIT SALE:	
	Transaction #:	4
	Card Type:	VISA
	Account:	8291
	Entry	Swipe
	Amount:	\$1.00
	Ref. Number:	88888888
	Auth Code:	000000
	Reponse:	APPROVED
I AGREE TO PAY ABOVE TOTAL AMOUNT ACCORDING TO CARD ISSUER AGREEMENT (MERCHANT AGREEMENT IF CREDIT VOUCHER)		
X..... SIGNATURE		

EMV Receipt Sample	Sample 1 08/06/2015 15:15:06 CREDIT SALE: Transaction #: 10 Card Type: AMEX Account: 1007 Entry Contactless Amount: \$6.01 Response: APPROVED TC: AE5822DF4672F521 TVR: 0080000000 AID: A000000025010403 TSI: A800 ATC: 0001 APPLAB: AMERICAN EXPRESS
	Sample 2 08/06/2015 15:40:17 CREDIT SALE: Transaction #: 21 Card Type: AMEX Account: 1007 Entry Chip Amount: \$30.00 Ref. Number: 521815146507 Auth Code: 772727 Response: APPROVAL TC: 2FFEF5396D4CF051 TVR: 0000008000 AID: A000000025010801 TSI: F800 ATC: 0002 APPLAB: AMERICAN EXPRESS PIN Verified

Sample 3	
08/06/2015	15:44:25
CREDIT SALE:	
Transaction #:	24
Card Type:	AMEX
Account:	1006
Entry	Chip
Amount:	\$31.00
Ref. Number:	521815146510
Auth Code:	773535
Response:	APPROVAL
TC:	CE6B6467C25C77B4
TVR:	0200008000
AID:	A000000025010801
TSI:	F800
ATC:	0002
APPLAB:	AMERICAN EXPRESS
I AGREE TO PAY ABOVE TOTAL AMOUNT ACCORDING TO CARD ISSUER AGREEMENT (MERCHANT AGREEMENT IF CREDIT VOUCHER)	
X.....	
SIGNATURE	
Sample 4	
08/06/2015	15:48:31
CREDIT SALE:	
Transaction #:	27
Card Type:	AMEX
Account:	1006
Entry	Chip Fall Back Swipe
Amount:	\$36.00
Ref. Number:	521815144915
Auth Code:	774353
Response:	APPROVAL
I AGREE TO PAY ABOVE TOTAL AMOUNT ACCORDING TO CARD ISSUER AGREEMENT (MERCHANT AGREEMENT IF CREDIT VOUCHER)	
X.....	
SIGNATURE	
Sample 5	

	08/06/2015 16:06:38 CREDIT SALE: Transaction #: 35 Card Type: AMEX Account: 1006 Entry Chip Amount: \$68.00 Ref. Number: 521815144925 Auth Code: 778011 Response: APPROVAL TC: CB37DC7D0C4392A5 TVR: 0000008000 AID: A000000025010801 TSI: F800 ATC: 0002 APPLAB: AMERICAN EXPRESS
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